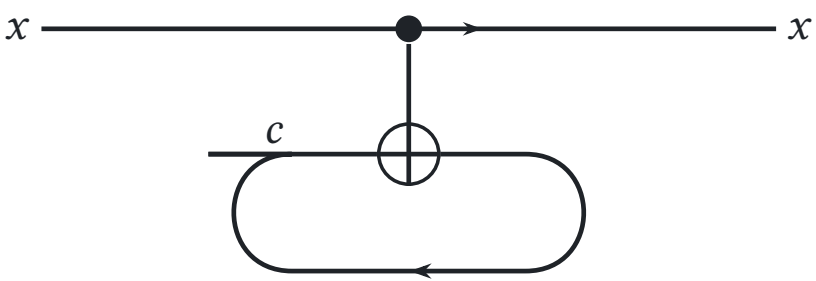
(a) trace for $\otimes$: a wire fed back beside the others



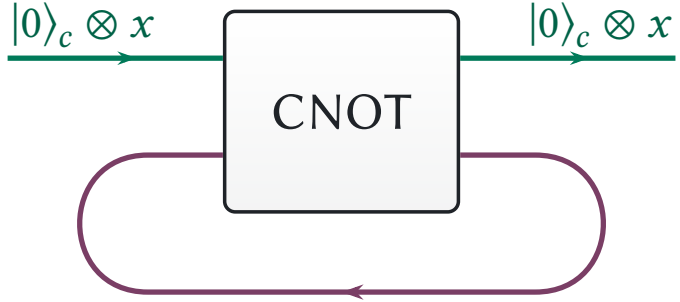
c goes back in beside x

$$\sum_k (I \otimes \langle k|)\,\mathrm{CNOT}\,(I \otimes |k\rangle)$$
$$= |0\rangle\langle 0|\,\mathrm{tr}(I) + |1\rangle\langle 1|\,\mathrm{tr}(\mathrm{X}) = 2|0\rangle\langle 0|$$

nothing splits and nothing passes twice;
not unitary

$$\mathrm{Tr}^{\mathbb{C}^2_c}_{\otimes}(\mathrm{CNOT}) : \mathbb{C}^2_x \longrightarrow \mathbb{C}^2_x$$

(b) trace for $\oplus$: the guard splits the state



$$A = |0\rangle\langle 0|,\ \ B = C = |1\rangle\langle 1|,\ \ D = |0\rangle\langle 0| \text{ on } x$$
$$\mathrm{Tr}^{\oplus}\,\mathrm{CNOT} = A + BC = I$$

$x = |1\rangle$ goes round once and leaves;

$$\mathrm{Tr}^{Y}_{\oplus}(U) : X \longrightarrow X \text{ unitary}$$

one controlled flip, x controls c; the guard decomposition reorders the tensor factors